

BASIC MECHANICAL ENGINEERING

UNIT-WISE MOST IMPORTANT TOPICS (PYQs BASED)

UNIT-I : Materials

Top 5 Most Important Topics

1. **Stress–Strain Diagram (Ductile & Brittle materials)**

Asked in: 2019, 2022, 2023, 2025

2. **Mechanical Properties (Strength, Hardness, Ductility, Toughness, Brittleness)**

Asked in: 2019, 2020, 2023, 2025

3. **Hardness Testing (Brinell, Purpose, BHN)**

Asked in: 2020, 2023, 2024

4. **Iron–Carbon Diagram**

Asked in: 2020, 2023

5. **Classification of Engineering Materials / Alloy Steels**

Asked in: 2019, 2022, 2024, 2025



Question Pattern

- Mostly **theory + diagram**
- Definitions + explain / short notes very common

UNIT-II : Measurement & Production Engineering

\Top 5 Most Important Topics

1. **Vernier Caliper (principle, use, least count)**

Asked in: **2019, 2020, 2022, 2023, 2024**

2. **Micrometer / Dial Gauge / Sine Bar**

Asked in: **2020, 2022, 2024, 2025**

3. **Errors in Measurement**

Asked in: **2020, 2023**

4. **Lathe Machine – Parts & Operations**

Asked in: **2019, 2023, 2025**

5. **Drilling Machine – Working & Operations**

Asked in: **2022, 2024, 2025**



Question Pattern

- **Instrument + neat sketch**
 - Short notes very frequent
-

UNIT–III : Fluids & Hydraulic Machines

Top 5 Most Important Topics

1. **Bernoulli's Equation (numerical + derivation)**

Asked in: **2019, 2020, 2022, 2025**

2. **Newton's Law of Viscosity**

Asked in: **2020, 2023, 2025**

3. **Types of Fluids / Newtonian vs Non-Newtonian**

Asked in: **2019, 2024**

4. **Pelton Wheel / Hydraulic Turbine (working)**

Asked in: **2019, 2023, 2025**

5. **Reciprocating Pump (working with diagram)**

Asked in: **2020, 2022**



Question Pattern

- Numerical + theory mix
 - Working principle with diagram compulsory
-

UNIT–IV : Thermodynamics & Steam Engineering

Top 5 Most Important Topics

1. Zeroth Law of Thermodynamics

Asked in: **2019, 2020, 2022, 2025**

2. First Law of Thermodynamics (statement + derivation)

Asked in: **2022, 2023, 2024, 2025**

3. Boilers – Classification & Working (Babcock/Wilcox, Cochran)

Asked in: **2020, 2023, 2024**

4. Boiler Mountings & Accessories (difference + function)

Asked in: **2019, 2020, 2022, 2024**

5. Natural & Artificial Draught

Asked in: **2020, 2025**



Question Pattern

- Long theory answers
 - Diagrams expected in boiler questions
-

UNIT–V : IC Engines, Cycles & Compressor

Top 5 Most Important Topics

1. Otto / Diesel / Dual Cycle (P-V & T-S diagrams + efficiency)

Asked in: **2019, 2022, 2023, 2024, 2025**

2. **4-Stroke Petrol & Diesel Engine (working)**

Asked in: **2019, 2020, 2023, 2025**

3. **2-Stroke vs 4-Stroke Engine (comparison)**

Asked in: **2019, 2025**

4. **Compressor (working principle)**

Asked in: **2020, 2025**

5. **Carnot Cycle**

Asked in: **2020, 2022**



Question Pattern

- **Diagram compulsory**
- Efficiency-based theory very common

PREDICTED MODEL QUESTION PAPER

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Attempt **any five** questions.
2. All questions carry **equal marks**.
3. Draw neat sketches wherever required.
4. Assume suitable data if necessary.

Q1 – MATERIALS (Very High Probability)

a) Explain the **stress-strain diagram for ductile material** with a neat sketch. Mark proportional limit, elastic limit, yield point, UTS and breaking point.

OR

b) Classify **engineering materials**. Explain **alloy steels** and write their important applications.

Q2 – MECHANICAL PROPERTIES & TESTING

a) Define the following mechanical properties (any four):

i) Strength ii) Hardness iii) Ductility iv) Toughness v) Brittleness

OR

b) Explain the **Brinell hardness test** with neat sketch. How is **BHN** calculated?

Q3 – MEASUREMENT & INSTRUMENTS (Sure Question)

a) Explain the **construction, principle and use of Vernier Caliper**. How is least count calculated?

OR

b) Explain the **working of Micrometer** with neat sketch and state its applications.

Q4 – PRODUCTION ENGINEERING

a) Explain the **main parts of a lathe machine** with neat sketch. Describe **any four lathe operations**.

OR

b) Explain the **construction and working of a drilling machine** and describe any two drilling operations.

Q5 – FLUID MECHANICS (Numerical + Theory)

a) State **Bernoulli's theorem**. Derive Bernoulli's equation and mention the assumptions made.

OR

b) The water flows through a pipe having diameters **25 cm and 10 cm** at sections 1 and 2 respectively. The rate of flow is **30 L/s**. Section 1 is **10 m above ground** and

section 2 is **2 m above ground**. If pressure at section 1 is **40 N/cm²**, find the pressure at section 2.

Q6 – HYDRAULIC MACHINES

a) Explain the **construction and working of a Pelton wheel turbine** with neat sketch.

OR

b) Explain the **working of a reciprocating pump** with neat diagram.

Q7 – THERMODYNAMICS & STEAM ENGINEERING

a) State and explain the **Zeroth law of thermodynamics**. Explain how it helps in temperature measurement.

OR

b) Explain the **construction and working of a Cochran boiler / Babcock & Wilcox boiler** with neat sketch.

Q8 – IC ENGINES & CYCLES (Very Repeated)

a) Explain the **working of four-stroke diesel engine** with neat sketch.

OR

b) Explain the **Otto cycle** with **P-V and T-S diagrams** and derive its efficiency.



WHY THIS PAPER IS IMPORTANT

- Almost **all questions are repeatedly asked** in PYQs
- Covers **every unit properly**
- Includes **sure topics**:
 - Stress–strain diagram
 - Vernier / Micrometer

- Bernoulli's theorem
 - Pelton turbine
 - Otto / IC engine
-

PREDICTED MODEL QUESTION PAPER – 2

Time: 3 Hours **Maximum Marks:** 70

Instructions:

1. Attempt **any five** questions.
 2. All questions carry **equal marks**.
 3. Draw neat and labelled diagrams wherever required.
 4. Assume suitable data if necessary.
-

Q1 – ENGINEERING MATERIALS

a) Explain the **classification of cast irons** along with their **applications**.

OR

b) Explain the **Iron–Carbon equilibrium diagram**. Mention different critical points and phases.

Q2 – MECHANICAL PROPERTIES & TESTING

a) Explain the **tensile testing of steel**. Draw and explain the **stress–strain curve for ductile material**.

OR

b) Define **Hooke's law**. Explain **Young's modulus** and derive its expression.

Q3 – MEASUREMENT & METROLOGY (Very Repeated)

a) Explain the **construction, principle and applications of Sine bar** with neat sketch.

OR

b) What are **errors in measurement**? Explain different types of errors and methods to minimize them.

Q4 – MACHINE TOOLS / PRODUCTION

a) Explain the **main parts of a lathe machine** with neat sketch. Explain **facing and knurling operations**.

OR

b) Explain the **process of sand casting**. Write its advantages and limitations.

Q5 – FLUID MECHANICS

a) State and explain **Newton's law of viscosity**. Differentiate between **Newtonian and Non-Newtonian fluids**.

OR

b) Derive **Bernoulli's equation** and explain its practical applications.

Q6 – HYDRAULIC MACHINES

a) Explain the **working principle of a Pelton wheel turbine** with neat sketch.

OR

b) What is a **draft tube**? Why is it used in reaction turbines? Explain with diagram.

Q7 – THERMODYNAMICS & STEAM ENGINEERING

a) State and explain the **First law of thermodynamics** for a **closed system**.

OR

b) Differentiate between **boiler mountings and boiler accessories**. Explain the working of **economizer**.

Q8 – IC ENGINES & CYCLES

a) Explain the **working of two-stroke petrol engine** with neat sketch.

OR

b) Explain the **Diesel cycle** with **P-V and T-S diagrams** and derive its thermal efficiency.
